



Review

Using artificial intelligence to rapidly identify microplastics pollution and predict microplastics environmental behaviors



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HIGHLIGHTS

- Artificial intelligence-driven methods for analyzing and predicting microplastics are reviewed.
- This review discusses data source and preprocessing for machine learning to predict microplastics.
- This review expounds various machine learning algorithm structures, pros and cons.
- This review analyzes the current limitations of machine learning to predict microplastics.
- This review discusses potential future directions for AI-driven microplastics prediction.

GRAPHICAL ABSTRACT



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ABSTRACT

With the massive release of microplastics (MPs) into the environment, research related to MPs is advancing rapidly. Effective research methods are necessary to identify the chemical composition, shape, distribution, and environmental impacts of MPs. In recent years, artificial intelligence (AI)-driven machine learning methods have demonstrated excellent performance in analyzing MPs in soil and water. This review provides a comprehensive overview of machine learning methods for the prediction of MPs for various tasks, and discusses in detail the data source, data preprocessing, algorithm principle, and algorithm limitation of applied machine learning. In addition, this review discusses the limitation of current machine learning methods for various task analysis in MPs along with future prospect. Finally, this review finds research potential in future work in building large

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generalized MPs datasets, designing high-performance but low-computational-complexity algorithms, and evaluating model interpretability.

1. Introduction

Soil, water, and sediment are all enriched with a large amount of microplastics (MPs, it means plastic particles with a size of less than 5 mm) [1,2], and the types of MPs are very diverse, including polyethylene (PE), polypropylene (PP), polyvinyl chloride (PVC), polystyrene (PS), polyurethane (PU), polyethylene terephthalate (PET), and polycarbonate (PC), etc [3]. The presence of MPs affects organisms in both water and soil, which poses a serious threat to the stability of ecosystems. Recently, a number of studies have shown that high concentrations (100 mg/L) of PS powder inhibited the growth of *Rheinheimera* bacteria by 45.8 % [4]. PE and polylactic acid caused up- or down-regulation of the abundance of various bacteria [2]. In addition, MPs act as carriers for other pollutants, exacerbating the toxic effects of other pollutants on organisms [5,6]. Given the potential harm of MPs to the ecosystem, it is necessary to investigate and count its distribution and behavior. Due to the complex and varied environmental media (present in oceans, land, atmosphere, and rivers [7]), there are also false positive MPs (Similar in appearance to MPs), such as microalgae, making qualitative and quantitative analysis of MPs particularly challenging [8]. The extensive distribution of MPs makes it impractical to use manual methods to investigate and quantify their concentrations, sizes, and distribution globally [9]. Traditional methods of investigating and counting MPs contamination usually involve spot sampling, followed by artificial identification and determination under a microscope. This

process may incorrectly define impurities such as microalgae and wood chips as MPs. High time and economic costs if instrumental analysis of the chemical composition of the sample is used.

In recent years, Artificial Intelligence (AI) has developed rapidly and has been widely used in areas such as the analysis and identification of MPs. In particular, machine learning (ML) comes under one direction of AI, and numerous algorithms of ML have been widely used in classification and regression prediction of MPs by virtue of their good fitting ability to continuous or discrete data. Meanwhile, driven by algorithmic advancement and computational power, deep learning (DL), which comes under ML, and focuses on neural networks inspired by the human brain, has been gradually applied to the prediction of MPs in the environment, especially in image processing, with remarkable results [10]. Although DL belongs to a sub-direction of ML, DL has a different mode of operation than traditional ML. Fig. 1 illustrates the main differences between traditional ML and DL. In traditional ML, human-engineered functions are designed to convert data into a shallow format suitable for traditional ML models, which are used to model and output the corresponding results. DL models autonomously learn feature representations of various data types (text, speech, images, video, etc.) to efficiently extract information and features from the data. The DL paradigm simplifies the entire process from the data to the final target output. It has been favored by a wide range of researchers. Therefore, this review discusses DL and traditional ML separately.

The use of ML and DL to assist in the investigation of MPs

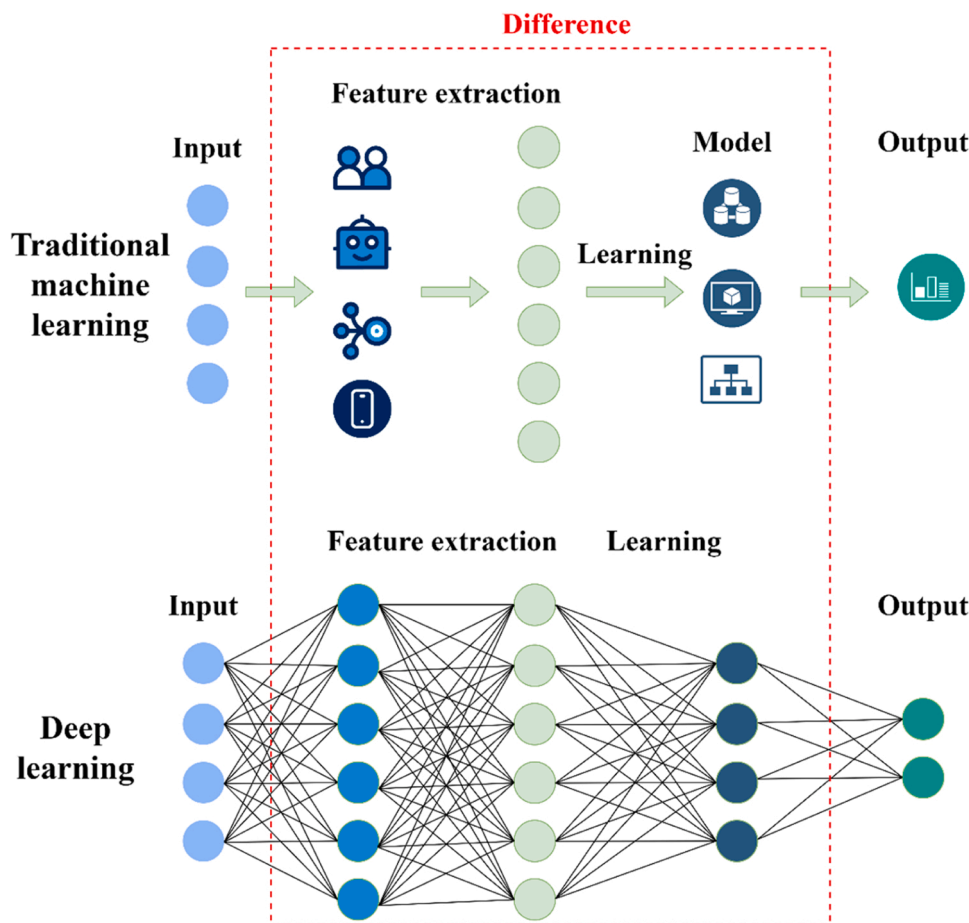


Fig. 1. Main difference between traditional machine learning and deep learning.

contamination can greatly improve work efficiency and reduce the probability of human judgment errors. Fig. 2 illustrates a count of the number of publications in the last five years where ML and DL have been applied to study the field of MPs (search tool, Web of Science; search date, May 12, 2024; “search topic, microplastics, deep learning” and “microplastics, machine learning”). ML published 206 studies, showing an upward trend year after year. Among them, 78 studies were published by DL, which is popular among researchers. This suggests that the use of AI is a feasible and efficient method to study the environmental pollution of MPs.

Currently, a small amount of literature reviews the application of ML to MPs [11,12]. Reference [11] (2023) focuses on key parameters such as the physical properties of MPs, while describing that image-based ML provides perspectives for MPs transportation. Reference [12] (2022) summarizes the progress of a small number of ML techniques in identifying MPs in environmental media. Both references emphasize the advantages of traditional ML methods for the transportation or identification of MPs. They lack a detailed overview of the data sources and preprocessing methods required for the modeling process of traditional ML and DL algorithms. In addition, they did not specifically explain the characteristics of each ML algorithm and did not provide an overview of DL applications. There is a lack of unified guidance on the use of AI techniques for classification, regression, image classification, image detection, and image segmentation of MPs.

To tackle these problems, this paper reviews AI techniques for predicting MPs with the aim of using AI to quickly identify MPs contamination and predict MPs environmental behavior. The main contributions of this study are as follows: (1) This paper comprehensively clarifies the data sources and data preprocessing methods used for ML prediction of MPs. (2) This paper discusses the structure of traditional ML and DL algorithms and the application scenarios suitable for them. (3) This paper discusses the current limiting limitations of ML methods applied to the analysis of various tasks in MPs, while proposing further ideas. (4) This paper finds high potential research value in constructing large generalized datasets of MPs, designing high-performance but low-computational-complexity algorithms, and evaluating the interpretability of models.

2. Data source and pre-processing

2.1. Data source

Data are crucial for model training. In particular, quality assurance and quality control (QA/QC) is essential in the data production process to prevent and minimize contamination during analysis [13]. Analytical

instruments were used to obtain chemically informative data and imaging pictures of MPs in samples, ML models can realize the detection of MPs by extracting the feature information in the data and recognizing the pictures [14]. Images primarily come from high-resolution digital cameras, digital microscopes with cameras, and scanning electron microscopes [15]. The data used for ML comes from a variety of instruments and methods. The sources of data are diverse. Considering that soil, and water are sources and sinks of MPs, current detection methods mainly focus on identifying MPs pollution in water, and soil [16]. We summarize the data source, as show in Fig. 3. Next, we will carefully describe the source of data for identifying MPs in water and soil and for predicting the environmental behavior of MPs, respectively. In addition, we introduce different data preprocessing methods for numerical data and image data to comply with model training and prediction according to different tasks.

Targeting MPs in water, with the most common methods including Fourier-transform infrared (FTIR) spectroscopy and Raman spectroscopy. Raman spectroscopy is a scattering spectroscopy where different MPs exhibit unique molecular vibration and rotation information [17]. However, the collected Raman signals contain a lot of interference, and the signal-to-noise ratio can be improved by the following methods: 1) merging multiple characteristic peaks using logic-based algorithms [18]; 2) decoding the spectral matrix using principal component analysis for identification [19]; 3) Pre-treatment of samples prior to Raman analysis reduces fluorescence interference from organic and inorganic substances. Pre-treatment methods include: reduction of organic matter using Fenton treatment [20], alcohol treatment [21], peroxide digestion (hydrogen peroxide digestion) [22], enzymatic maceration according [23], etc.; and reduction of inorganic interferences using filtration, magnetic separation, and density separation [24]. Raman identification of MPs has good size resolution and can identify MPs particles below 1 μm in size [25]. Due to the different absorption spectra of common MPs, analysis of absorption spectra can identify and quantify MPs. Therefore, fourier transform infrared spectroscopy is also commonly used to detect MPs pollution in water [26]. For MPs particles larger than 20 μm , FTIR provides more reliable data compared to Raman spectroscopy [27]. When identifying MPs in aquatic environments, Raman spectroscopy is not affected by water signal interference, unlike FTIR, which cannot avoid this issue [25]. In addition to these two most common methods, methods for detecting MPs in water are quite plentiful. For example, staining MPs and then using flow cytometry to detect them in water samples [28]. This method can identify even smaller MPs particles, with spatial detection limits as low as 200 nm [29]. Another characteristic of flow cytometry is that it not only extracts parameters of MPs for quantification but also provides richer morphological

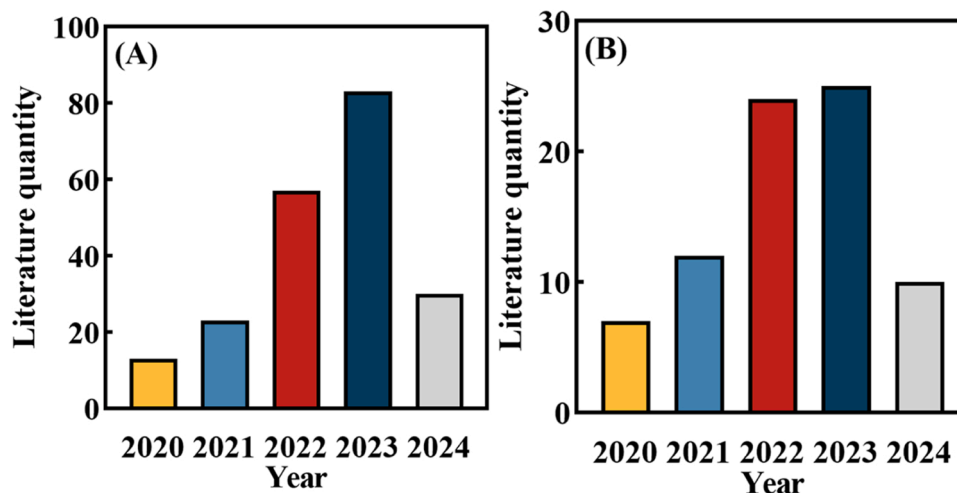


Fig. 2. (A) Number of literatures of ML in MPs. (B) Number of literatures of DL in MPs.

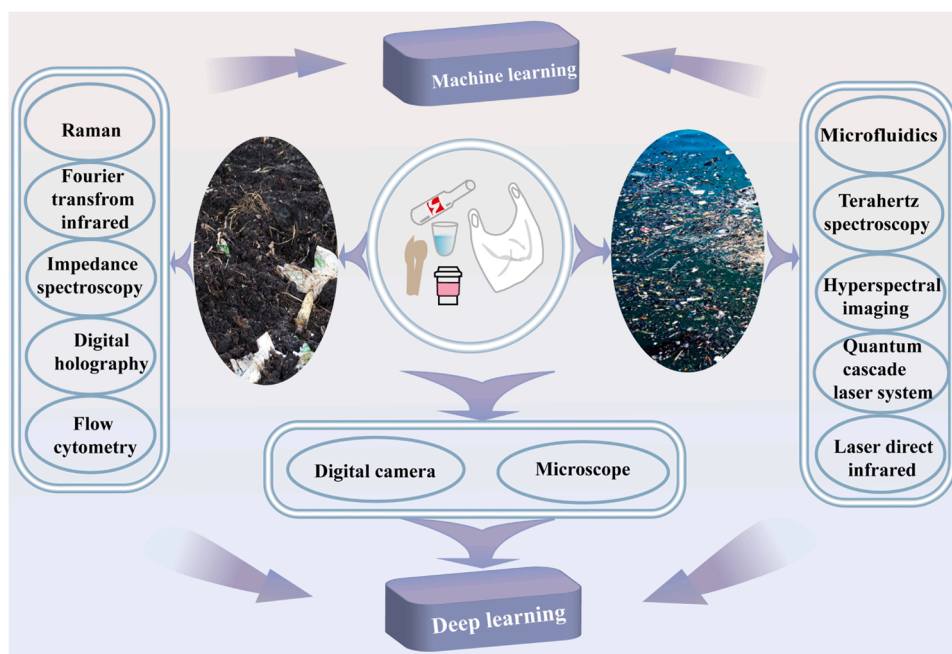


Fig. 3. Summary of data source.

information, such as curvature, straightness, roundness, length, and thickness of MPs [30]. MPs of different shapes and chemical properties exhibit differences in polarized light upon incidence, making polarized light optical microscopy effective for identifying and counting MPs in wastewater [31]. In particular, Wang et al. (2018) [32] found that polarized light scattering at 120 °C is sensitive to particles and can effectively differentiate MPs particles on a large scale. Digital holography is an in-situ detection technology that enables high-throughput sample detection and comparative imaging without sample destruction [33]. In aquatic environments, digital holographic microscopy can effectively distinguish MPs from microorganisms at the micrometer level, such as diatoms [34]. Impedance spectroscopy is a rapid identification method for MPs larger than 300 μm , where the size and chemical composition of MPs affect resistance and impede current flow, careful analysis of this information allows for MPs identification [35].

Targeting MPs in soil, hyperspectral imaging technology has been used for identifying MPs. Because of the carbon-hydrogen bonds (C-H) of MPs cause spectral vibrations, hyperspectral imaging using near-infrared spectroscopy can directly analyze and quantify MPs in soil without the need for extraction [36]. Hyperspectral imaging technology can also differentiate between soil with abundant leaves, rocks, and branches [37]. Furthermore, hyperspectral imaging technology can obtain spatial distribution characteristics of MPs in surface soil [38]. Compared to hyperspectral in situ measurement techniques, other methods require some pretreatment of the sample, including Laser Direct Infrared Chemical Imaging System [39,40], Terahertz Spectroscopy [41], and Microfluidics [35] [42], Gas Chromatography-Mass Spectrometer (GC-MS) [43].

ML are not only widely applied in MPs pollution identification but also extensively studied for predicting their environmental behaviors. For example, the prediction of pollutant adsorption by MPs [44], release projections for plastics [45], prediction of toxic effects of MPs on organisms [46]. The data in this area come from diverse sources depending on specific research scenarios. ML is trained on these datasets for the purpose of predicting the behavior of MPs in a certain environment. Among them, the adsorption behavior of MPs towards pollutants, with data primarily sourced from adsorption kinetics and thermodynamics experiments [47]. Adsorption kinetics data include adsorption equilibrium time and rate, while thermodynamic parameters involve the

relationship between adsorbent and pollutant concentrations upon reaching adsorption equilibrium [48]. Factors affecting adsorption may include pollutant concentration and type [49], type and amounts of MPs [50], solution pH, ion strength, salinity, and environmental temperature [51]. These are potential factors that affect the prediction results and will be used as independent variables in the dataset when performing model training.

2.2. Data pre-processing

It is necessary to extract the diversity data of MPs using relevant instruments and feed them into the model for training after corresponding preprocessing, to ensure the prediction accuracy and generalization ability of the model. There are usually two types of MPs data, numerical data and image data, and we will explore the preprocessing methods for the two types of data separately.

Numerical data, such as data extracted after FTIR, Raman spectroscopy, some factors in the experiment (time, temperature, instrument class, etc.), are usually used for classification and regression tasks. In order to ensure that the model can correctly and efficiently understand the feature connections that exist between the data, the data need to be preprocessed before model training. For example, 1) Maximum value normalization [52], which linearly maps the feature values to the 0–1 interval through the sample feature maximum values. 2) Category balancing [53], which expands or deletes data from samples with large differences in the number of categories in order to achieve category balancing. 3) Missing value processing [54], which will infer the missing values based on the distributions of the existing data, and is usually used as a mean-padding or a plurality-padding. 4) One-hot coding [55], which is used when the feature information is the category feature. When the feature information is a category feature, one-hot coding is usually used to encode the category (e.g. 00, 01, 10, 11 represent 4 categories respectively). 5) Data dimensionality reduction [56], to delete some highly similar dimensional information to achieve better model fitting.

Image data, for MPs, usually requires high-quality preprocessing before it can be fed into a DL model for training to achieve high-performance analysis results. Usually, low resolution images are manually removed, and then python (<https://www.python.org/>) is used to perform image quantity enhancement operations (e.g., vertical

rotation, random rotation, panning, scaling, etc.) to expand the dataset and prevent model overfitting. For the object detection task, the LabelImg (<https://github.com/tzutalin/labelImg>) tool is used for data labeling. For image segmentation task, LabelMe (<https://github.com/wkentaro/labelme>) tool is used for labeling. The processed image dataset and labels are fed into the model for model training and prediction.

3. Overview of traditional machine learning algorithms

Traditional ML methods provide a reliable solution for diversity data assessment of environmental and ecological risks of MPs. Traditional numerical models relying on mathematical equations or empirical methods can be both time-consuming and inefficient when dealing with diversity data or complex systems [57]. Data-driven ML approaches can accelerate the control of harmful MPs and reduce the risk of MPs in localized areas or even globally [58] [14]. Their applications are shown in Table 1 ML methods have been widely used in various environments to predict their environmental behavior. They have even achieved more than 90 % accuracy (such as [60–63] [67]) and 90 % R^2 value (such as [64,65] [69,70]) in a variety of tasks. Therefore, ML models have demonstrated strong learning ability and high prediction accuracy in modeling complex environmental systems.

3.1. Traditional machine learning classification

ML classification algorithms have been widely applied in the analysis of MPs [73], such as decision tree (DT) [74], random forest (RF) [75], support vector machine (SVM) [62,67], k-nearest neighbors (KNN) [68], etc. These ML classification algorithms aim to extract feature information between MPs data samples, classify and identify MPs based on the feature information, reduce labor costs, and improve the accuracy of MPs category information [76].

3.1.1. Decision tree (DT)

Decision tree is a model for presenting decision rules and

classification results in a tree data structure [77,78]. In the DT generation algorithm, the information gain of each feature is calculated initially. Subsequently, the feature with the highest information gain is selected. Then, based on this selection, the information gain of other features is calculated, and the feature with the highest information gain is chosen. Once all features have been selected, the decision tree for predicting MP type is obtained [79]. DT classification algorithm using real MPs classification research demonstration. Carrera et al. (2022) [80] used near infrared spectroscopy (NIR) (wavelength range between 0.8 and 2.5 μm) to identify four polymer classes, PE, PET, PP, PVC. The dataset was acquired by a handheld spectrometer called MicroNIR site and the datasets were pre-processed and normalized. The wavelength range of the data was taken as 1000 ~ 1600 nm. The structure of the DT classification algorithm, as shown in Fig. 4(A), the positions of two sets of characteristic absorption peaks were extracted for each MPs class and the decision tree rules were formulated based on the peak positions. When the absorption peaks belonged to 1204 nm and 1386 nm, the substance was determined to be PE; when the absorption peaks belonged to 1118 nm and 1405 nm, the substance was determined to be PET; when the absorption peaks belonged to 1185 nm and 1386 nm, the substance was determined to be PP; and when the absorption peaks belonged to the other positions, the substance was determined to be PVC. DT employs a tree data structure to represent decision rules and classification outcomes. Serving as an inductive learning algorithm, it aims to convert seemingly disordered and chaotic known data into a tree model, enabling the prediction of unknown data through specific technical methods. Each path from the root node (the attribute contributing most to the final classification) to the leaf node (the ultimate classification outcome) represents a decision rule. Based on multiple decision rules, the category to which the input data belongs can be determined. Ai et al. (2022) [76] and Xu et al. (2023) [81] multiple hyperspectral data of agricultural soil samples were collected using a hyperspectral imaging spectrometer. Principal component analysis was employed to minimize the dimensionality of the hyperspectral data. The collected hyperspectral data underwent preprocessing steps including Savitzky-Golay smoothing filtering, mean standardization, and principal

Table 1
Application of machine learning models.

MPs source	Objective	Data analysis	Task	Model	Performance	Reference
Rubber particles	MPs counting	FTIR	Regression	LR, DTR	LR>DTR, MAE (LR)< 6.3 %	[59]
MPs in water	Distinguish between more than 20 polymers	FTIR	Classification	RF	Accuracy= 97 %	[60]
Tap water nanoplastics	Distinguishes nanoplastics from other particles	Raman	Classification	RF, SVM	RF>SVM, Accuracy (RF)= 98 %	[61]
Wastewater treated by a bio-filter	Identify the 11 most common MPs	FTIR	Classification	RF	Accuracy= 98 %	[62]
Tap water MPs	Detection of water samples contaminated by different concentrations of MPs	EIS	Regression, Classification	SVM	Classification accuracy= 99.2 %, Regression accuracy= 96 %	[63]
Various natural aging environment MPs	The adsorption capacity of care products for long-term aging MPs	FTIR	Regression	DTR	$R^2 = 97 \%$	[64]
MPs in soil	Effect of MPs in soil P, pH, and doc	-	Regression	GBR, DTR, RFR	GBR>DTR>RFR, R^2 (GBR)= 86–99 %	[65]
MPs in urban soil of Nanjing	Growing trend of MPs in urban soil	FTIR	Regression	DTR	Accuracy= 79.8 %	[66]
Plastic products	Characterization of the composition of MPs in plastic products	Raman	Classification	SVM	Accuracy= 96 %	[67]
Household plastic products	Analysis of MPs for household plastic products	Raman	Classification	SVM, KNN	Accuracy (SVM)> 88 %	[68]
Various water sources	Modeling batch adsorption capacity of methylene blue on MPs	FTIR	Regression	RFR	$R^2 = 97.55 \%$	[69]
MPs in soil-cucumber system	Effect and mechanism of MPs contamination in soil-cucumber system	-	Regression	RFR	$R^2 = 94 \%$, RMSE= 2.41	[70]
MPs in marine sediments	Classifying MPs in marine sediments (e.g., sand)	NIR-HSI	Classification	SVM	High accuracy	[71]
MPs in peatland sediment	The potential for predicting the quantities of MPs in peatland sediment	FTIR	Regression	RF, LS - SVM, LSTM	LSTM>RF>LS-SVM, R^2 (LSTM)= 68 %	[72]

Remark: FTIR- Fourier-transform infrared spectroscopy; Raman- Raman spectroscopy; EIS- electrical impedance spectroscopy; NIR-HIS- near infrared hyperspectral imaging; MAE- mean absolute error; R^2 - R-Square; RMSE- root mean square error

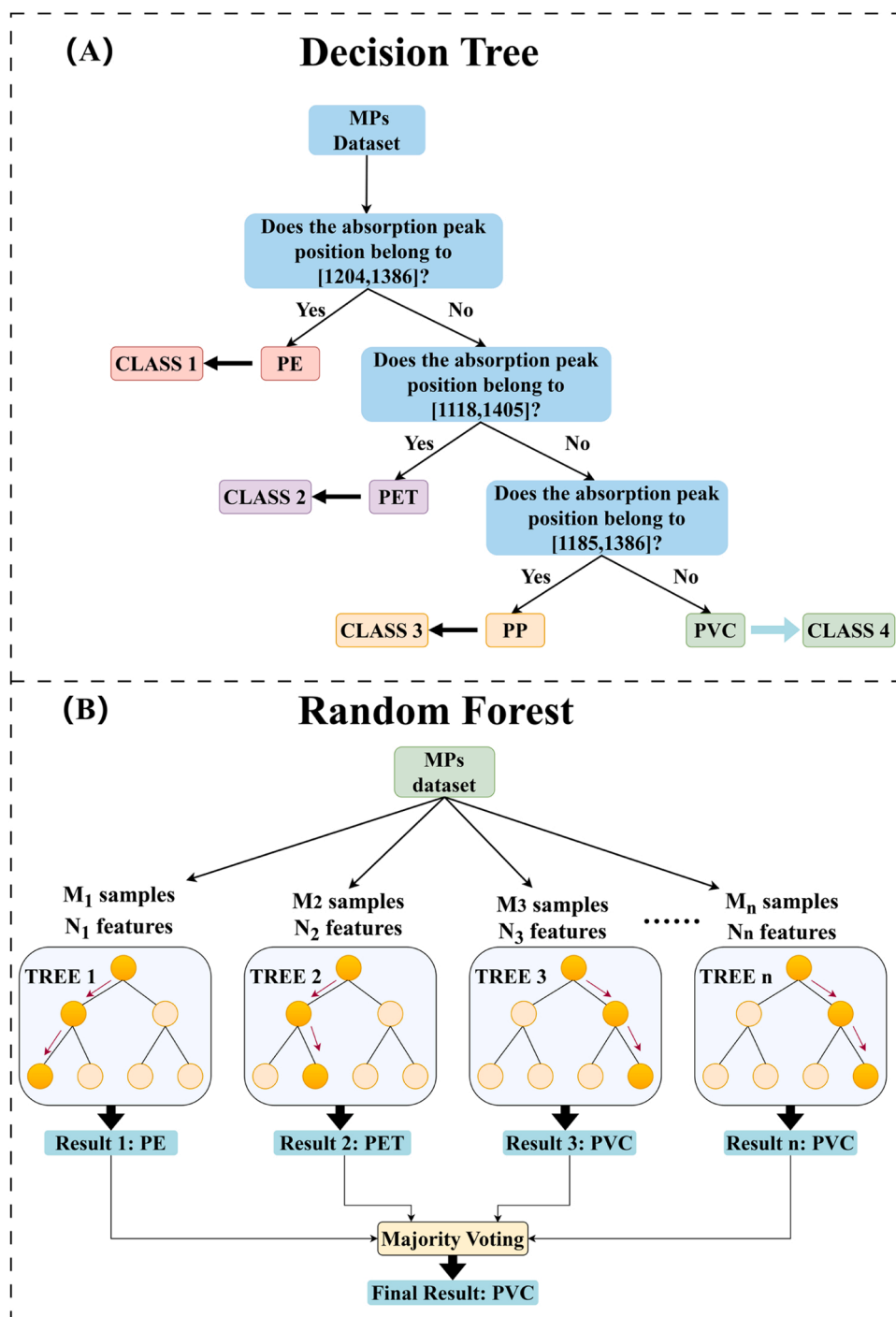


Fig. 4. Algorithm structure schematic. (A) Structure diagram of the DT classification algorithm. Taking near-infrared spectra as an example, the category of four common MPs is determined based on absorption peaks in the NIR range of 1000 nm to 1600 nm; (B) Structure diagram of the RF classification algorithm.

component analysis for dimensionality reduction. Finally, a DT model was built to classify three types of polymer materials (PE, PP, PVC) present in agricultural soil.

DT can handle both continuous and discrete variable data and achieve fast processing speed. However, DT are prone to overfitting and low generalization ability during training. Such problems can usually be solved by setting the maximum depth of the tree.

3.1.2. Random forest (RF)

Random forest is derived from ensemble learning through multiple DT algorithms [82]. It consists of multiple decision trees, and each DT

model is trained on a random subset of the original dataset, obtained through sampling with replacement [83]. Subsequently, the predictions of all models are combined through a voting mechanism. From an intuitive perspective, each DT acts as a classifier, producing a classification result for a given input sample. In the RF classification algorithm, all the categorical votes are aggregated and the category with the most votes is used as the final output. In addition to random subsets of the dataset, each DT also selects the best split from a random subset at each node. This additional randomness helps reduce correlations between models, further improving model performance [84]. We plotted the RF classification algorithm, as shown in Fig. 4(B). The number of votes for

the three MPs PE, PET, PVC is 1:1:2 and hence the final classification result is PVC. The workflow of RF is as follows: 1) Randomly selecting samples: N samples are randomly chosen from the original training dataset (typically with replacement), meaning the same sample may be selected multiple times. Each decision tree is trained using these randomly selected samples. 2) Randomly selecting features: at each split point of the DT, the algorithm randomly selects a subset of features and identifies the best splitting feature from this subset. This approach increases model diversity and reduces the risk of overfitting. 3) Building DT: for each DT, the best splitting feature is first selected, the node is split based on this feature, forming branches of the tree, and then each branch forms a new node, repeating this process. 4) Training multiple DT: the above steps are independently carried out on each tree, where each tree has different combinations of samples and features, making each tree unique. 5) Summarizing prediction results: in classification

problems, the "majority voting" principle is used, meaning each tree provides a prediction category, and the final prediction result is the category with the most votes.

RF, with excellent performance such as strong generalization, fast training, and strong handling of high-dimensional data [85–87], has been widely applied by many scholars in the classification of MPs, achieving very good results. Mao et al. (2024) [88] quantified the relative importance of each influencing factor on the MP removal efficiency using RF modeling, which results are ordered as follows: MPs shape (28.1 %) > coagulant type (14.3 %) > coagulant dosage (14.0 %) > MPs concentration (12.0 %) > MPs size (11.9 %) > MPs type (11.4 %) > pH (8.3 %). Valls-Conesa et al. (2023) [62], extracted training and testing spectra from FTIR spectra images of pure MPs samples and automated the process using reference spectra, fast background correction, and identification algorithms. They built a RF

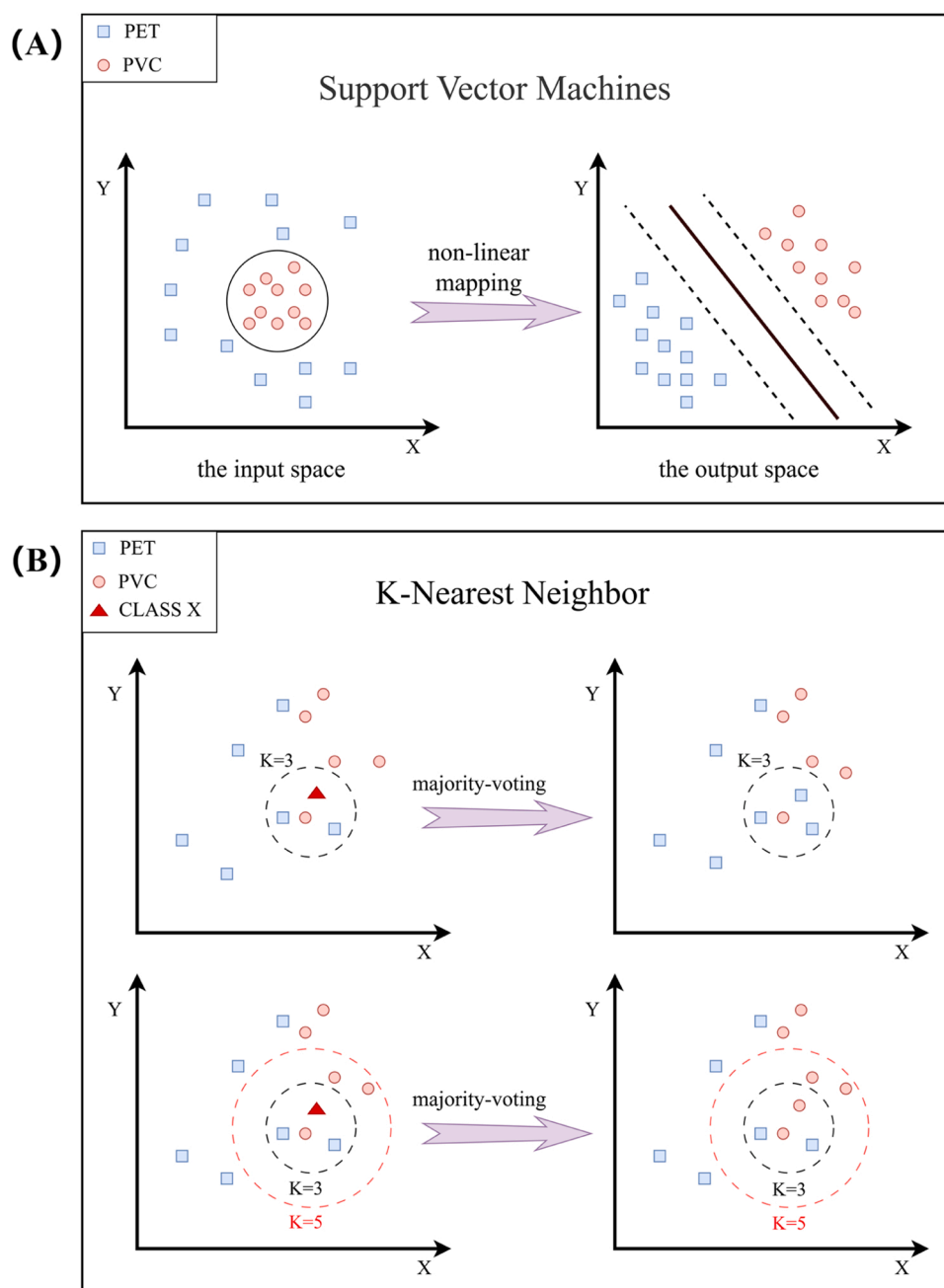


Fig. 5. Algorithm structure schematic. (A) Structure diagram of the SVM classification algorithm; (B) Schematic diagram of the KNN classification algorithm. X-axis and Y-axis both represent the feature information of the MPs, such as X for color, size, hardness, etc., and Y for Raman spectral value, infrared spectral value, etc.

classification model validated against program-generated ground truth to rapidly identify 11 common types of MPs in FTIR spectra from the environment. Hufnagl et al. (2019) [75], proposed a method to differentiate different polymer types, measure their abundance and size distribution, and applied a RF classifier to calculate a multi-class model for polymers including PE, PP, polymethyl methacrylate (PMMA), polyacrylonitrile (PAN), and PS.

RF, although strong for high-dimensional data processing ability and fast training speed, however, require a large amount of data. For small or low-dimensional data, it may not produce good classification results. Therefore, a large amount of data is necessary for RF.

3.1.3. Support vector machine (SVM)

The support vector machine classification model is a linear classifier defined in feature space with the maximum margin and has demonstrated its effectiveness in many MPs classification problems [89], obtaining better performance than other classification algorithm models [90–92]. SVM maximizes the margins of the training dataset by finding an optimal separating hyperplane [93]. Fig. 5(A) illustrates the SVM model with a linear kernel mapping the nonlinear input space to a new linear separable space [94]. Specifically, all vectors on one side of the hyperplane are labeled as -1 and all vectors on the other side are labeled as $+1$. The training instances that are closest to the hyperplane in the transformation space are called support vectors. The number of these support vectors is usually small compared to the training set. As shown in Fig. 5(A), each point represents a single MPs sample, and different colors represent different MPs categories. X-axis and Y-axis both represent the feature information of the MPs, such as X for color, size, hardness, etc., and Y for Raman spectral value, infrared spectral value, etc. Each MPs sample is composed of multiple feature coordinates based on its own X and Y values $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, each of feature coordinate represents one MPs sample distributed in two dimensional coordinates. Finally, a scatter plot is obtained as shown in the input space. These data points are distributed in a nonlinear space and appear disorderly. Even an approximate linear division cannot be found among the training data, making it impossible to effectively separate positive and negative classes. Therefore, it is necessary to map this nonlinear input space into a new linearly separable space to achieve better classification results. V et al (2023) [63] used an SVM classifier to classify MPs by measuring the electrical impedance spectroscopy of MPs in tap water with an accuracy of more than 98 % for PP, PET, PVC, PC, and PU, and more than 70 % for high-density PE and low-density PE. Feng et al. (2023) [68] proposed a household MPs classification multi-model ML method based on Raman spectra. SVM was used to accurately identify seven standard MPs samples, real MPs samples, and MPs samples exposed to environmental stress, real MPs samples, and real MPs samples exposed to environmental stress, providing a valuable MPs classification tool by coupling classification models with Raman spectra.

SVM can solve the classification of nonlinear data and has a strong fit to small sample data. It cannot fully show its performance for large-scale data and missing data. Therefore, the adoption of SVM requires the appropriate dataset without missing data.

3.1.4. K-nearest neighbor (KNN)

K-nearest neighbor is one of the simpler classification algorithms and also one of the most commonly used supervised classification algorithms [95]. KNN calculates the distance between the unknown category and all known categories. The K nearest known samples is selected based on the calculated distances. Majority voting rule are used to categorize unknown categories into categories with K number of nearest neighbor categories [96]. Fig. 5(B) shows a schematic diagram of the KNN classification algorithm. Two types of MPs, PET and PVC, are distributed along the coordinate axes based on their feature information. Now, we need to determine the type of a MPs located at a certain position (denoted as X). We choose an appropriate value of K and use the

categories of the K nearest points to X to determine which class X belongs to. As shown in Fig. 5(B), when $K=3$, we select the closest 3 points to X and find that two points are PET and one point is PVC. We conclude the MPs to PET by voting. However, when $K=5$, the classification changes. Among the closest 5 points to X, 3 points belong to PVC and 2 points belong to PET. Therefore, we determine that this MPs belongs to the category of PVC by voting. Therefore, the choice of K value is crucial for the KNN classification. Cross-validation (splitting the sample data into training and validation data in different proportions) is often used by starting with a small K value and gradually increasing the K. The variance of the validation set is calculated to find the right K value. The KNN classification algorithm has a simple structure, and it is easy to understand and shows high performance without parameter estimation or training, is commonly used by scholars for MPs classification research. Kedzierski et al. (2019) [97] established a learning database consisting of 969 MPs spectra and applied the KNN classification method to MPs spectra collected from the Mediterranean Sea by Tara. The proposed automated method yielded results within 10 % difference compared to human expert knowledge. Yan et al. (2022) [91] proposed an ensemble learning algorithm that systematically evaluates existing ML-based methods for MPs identification, taking advantage of different ML models. KNN integrates the output of multiple ML algorithms to improve classification accuracy, which is superior to existing single-ML methods.

KNN has the advantages of high accuracy, insensitivity to outliers, etc. It is also accompanied by the problems of high computational complexity and high spatial complexity. In practical applications, suitable algorithm parameters are chosen based on the actual situation to achieve the best classification results.

ML classification models avoid cumbersome manual judgments, while demonstrating high performance against customized datasets and making appropriate judgments when encountering new samples [98]. These traditional ML classification algorithms are adapted to a variety of application scenarios, achieving powerful feature extraction capabilities along with high prediction accuracy. Depending on different environmental domains and different data feature information, different ML models can be selected to improve the prediction performance. This concept can easily be applied to analyze other types of environmental particles, such as beach MPs [99], liquid-solid frictional triboelectric nanogenerator MPs [100], seafood MPs [101], etc., for better application in various downstream tasks.

3.2. Traditional machine learning regression

The classification technology of ML has achieved great success in many tasks of MPs. However, the classification technology is limited to the identification of categories of MPs and cannot meet the requirements for some specific tasks. Such as predicting the concentration of low-density PE, PET, and acrylonitrile-butadiene-styrene in beach sediments based on spectral reflectance [99], predicting the physical and chemical properties, weathering status, and concentration of MPs in water [102], identifying the abundance of MPs in soil environments [103], predicting the number of MPs particles [59], etc. For specific tasks with output results as specific numerical values, regression analysis of various types of MPs data is required. Regression model accuracy validation of MPs data is necessary for specific tasks that output results with specific values, such as mean absolute error (MAE), R-Square (R^2), root mean square error (RMSE) [72]. The basis of regression algorithms lies in their ability to model the relationship between features and targets, which enables applications such as prediction, relationship analysis, and feature importance assessment. Through regression analysis, we can understand trends between variables, detect outliers, provide support for decision-making, and conduct trend analysis, providing powerful tools for data science and practical problems [104].

3.2.1. Linear regression

Linear regression is a common statistical method used to analyze the relationship between variables [105], particularly for predicting the relationship between a variable and one or more independent variables [106]. One independent variable is called simple linear regression, and multiple independent variables are called multiple linear regression. As shown in Eq. (1): Y is the dependent variable (the variable to be predicted), X is the independent variable (the variable used for prediction), β_0 is the intercept term (the value of the model when $X=0$), β_n is the slope (indicating the effect of X 's change on Y), and ϵ is the error term. When X_2 to X_n terms are 0, it becomes a simple linear regression.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon \quad (1)$$

Linear regression is easy to understand and implement. Da Silva et al. (2020) [107] utilized this algorithm to provide an efficient automated method for characterizing MPs polymer, abundance calculation, and size distribution. Xu et al. (2021) [108] obtained comprehensive experimental data for the adsorption of 18 different kinds of neutral organic compounds on emerging polar MPs and conventional nonpolar MPs, a multiple linear regression model was used to accurately describe the overall adsorption of organic compounds on different MPs by different intermolecular interactions. Jia et al. (2024) [109] analyzed based on a linear regression model showing that PE and PVC MPs highly increased plant functions, while posed negative effects on soil functions. PS and biodegradable (Bio) MPs decreased plant functions, while stimulating soil functions. Additionally, low-density polyethylene (LDPE), PE, PS, PVC, Bio, and granular MPs significantly decreased soil microbial functions.

Linear regression is easy to understand and implement, and performs well when data linearity is strong. However, linear regression is sensitive to outliers and noise and only captures linear relationships and cannot handle nonlinear relationships. Therefore, linear regression is adapted to situations where the data presents linear relationships and has high requirements for interpretability.

3.2.2. Decision tree regression

Linear regression algorithms often exhibit poor predictive performance when data exhibit non-linear relationships and are sensitive to outliers and noise [110]. Decision tree regression is a regression model based on a tree structure. It recursively partitions the input space and takes the average of the target values of samples within each partition as the predicted value.

$$\hat{y}_m = \frac{1}{N_m} \sum_{i \in D_m} y_i \quad (2)$$

Where D_m represents the set of sample indices within the m -th leaf node, and N_m represents the number of samples within that leaf node. The decision tree regression algorithm is easy to understand and interpret, with good visualization effects. It can handle non-linear relationships and is insensitive to outliers and missing values. The decision tree regression algorithm is widely used in various fields, such as predicting the quantity of MPs particles [59,111], the adsorption capacity of pharmaceuticals and personal care products on MPs [64], and the growth trend of MPs in urban soil [66].

Decision tree regression is able to handle nonlinear relationships and is insensitive to outliers and missing values, but is less stable. Data with nonlinear relationships are usually used to train decision tree regression algorithms.

3.2.3. Random forest regression

Random forest regression is an algorithm for ensemble learning by integrating multiple decision tree regression algorithms. It improves the generalization performance of the model and reduces overfitting by building multiple decision trees and averaging or voting on them. It performs well on high-dimensional and large-scale data and can handle

non-linear relationships and complex feature interactions [112].

$$\hat{Y} = \frac{1}{N_{trees}} \sum_{i=1}^{N_{trees}} f_i(X) \quad (3)$$

Where N_{trees} is the number of trees in the forest, $f_i(X)$ is the prediction of the i -th tree. Random Forest regression is suitable for solving complex regression problems in MPs prediction, high-dimensional datasets, or issues such as noise and non-linear relationships in MPs datasets. Li et al. (2023) [113] utilized 15 environmental variables represent natural and anthropogenic environmental features, with 7 environmental variables selected for the experimental model. The Random Forest model fits the sample data, aiding in identifying regional MPs distribution. Using 2226 paired field/experimental measurements and employing a randomized forest regression algorithm, Gao et al. (2022) [114] found that crop growth and soil health responses to macro-and/or MPs were correlated with plastic content, plastic type, plastic size, and crop/animal type variables.

The random forest regression model has some drawbacks such as poor interpretability and possible overfitting on some datasets. We usually use it to solve problems such as datasets with high feature dimensions and the presence of noise and nonlinear relationships in the dataset.

3.2.4. Gradient boosting regression

Gradient boosting regression is a special "Boosting" algorithm that attempts to iteratively train weak learners in sequence to learn from mistakes and develop robust models [115], thus achieving better learning performance than individual weak learners [116]. The model minimizes the loss function at each iteration and learns from the experience of previous mistakes. The final model will be a collection of all weak learners, summarizing all predictions of each model and capable of handling complex nonlinear feature relationships in MPs. Withana et al. (2024) [65] employed gradient boosting regression for quantitative analysis of the impact of MPs on soil properties, revealing the effects of MPs on soil total phosphorus, pH, and dissolved organic carbon, contributing to understanding the role of MPs in soil systems.

Gradient boosting regression can handle complex nonlinear relationships. Gradually optimizes the model during the training process and gradually reduces the residuals. However, it has a relatively long training time, especially when the depth of the tree is large. Gradient boosting regression is suitable for solving nonlinear regression problems, as well as problems that contain noise and complex feature interactions in the dataset.

3.2.5. Support vector regression

Support vector regression is a method that uses SVM for regression. Support vector regression maps inputs from a low-dimensional space to a high-dimensional feature space, and finds the optimal hyperplane in the high-dimensional space for regression prediction [117]. Similar to SVM in classification tasks, support vector regression also employs kernel methods to map the input space to a high-dimensional feature space. The objective of support vector regression is to find a hyperplane that minimizes the difference between the mapped values of the training samples on this hyperplane and the target values while maintaining the error within a certain range [118]. Support vector regression performs well in high-dimensional spaces and is suitable for complex nonlinear regression problems in MPs. By adjusting the kernel function and parameters, it can adapt to different types of MPs data distributions. For example, suitability assessment of predicting MPs degradation migration pollutants [119], and prediction of MPs in ash based on NIR [120], and label-free identification of small MPs in seawater [98].

Support vector regression performs better in high-dimensional spaces, but for large-scale datasets, the training time is long, and it is sensitive to the scaling of the input data and the choice of parameters. Support vector regression is often used to solve nonlinear regression

problems, especially when it makes sense to map the input space to a high-dimensional feature space.

4. Overview of deep learning algorithms

Numerous classification and regression algorithms of ML have been applied to various predictions of MPs. However, ML algorithms can only perform tasks involving discrete data with multiple features of MPs. For specific visual tasks such as automatic counting [121,122], shape classification [121], and automatic detection of MPs [123], ML algorithms have shown significant limitations. DL is a new research direction in the field of ML, which has been introduced into ML to bring it closer to its original goal of artificial intelligence (AI). DL is the process of learning the intrinsic regularities and representation levels of sample data, and the information gained during this learning process can be of great help in interpreting data such as text, images, and sounds, etc. DL not only handles data tasks but also processes image files. The application of DL techniques in the field of MPs has become an irreversible trend. Its

applications are shown in Table 2 DL technology is applied in different MPs scenarios such as seawater [15,98], farmland soil [81], industrial wastewater [124], daily products [125,127] etc. It also demonstrated more than 90 % accuracy in a variety of tasks (such as classification, object detection, image classification, image segmentation) [76,81,98,121,123,124,126–130,132]. With its unique structure, under equivalent environmental factors, DL has shown better prediction performance than traditional ML in numerous applications (such as [76,81,98,128]).

Remark: LDIR-laser infrared imaging spectrometer; SERS-surface-enhanced Raman spectroscopy; SEM- scanning electron microscopy.

4.1. Convolutional neural network classification and regression

DL exhibits better prediction accuracy than ML for classification and regression of MPs under some specific tasks (such as [76,81,98,128]). Convolutional neural networks (CNN) belong to a specific method of DL, consist primarily of convolutional, pooling, and fully connected layers. The multilayered model structure of CNN allows them to automatically

Table 2
Application of deep learning models.

MPs source	Objective	Data process	Task	Model	Performance	Reference
MPs in seawater	Detection of microzooplankton and MPs in the ocean	Camera	Object detection	YOLOV5	Accuracy > 80 %	[15]
MPs in the soil	Classification of three polymers, PE, PP and PVC, in agricultural soil	Hyperspectral Imager	Classification	DT, SVM, CNN	Accuracy (CNN)= 92.6 %, Accuracy (DT)= 87.9 %, Accuracy (SVM)= 85.6 %	[76]
MPs in farmland soil	Classification of the three MPs polymers PE, PP and PVC and their combinations	LDIR	Classification	CNN, SVM	Accuracy (CNN)= 95 %, Accuracy (SVM)= 93.85 %	[81]
MPs in seawater	Identify 11 common MPs in seawater	Raman	Classification	SVM, RF, CNN, ResNet34	Accuracy (CNN)= 93.2 %, Accuracy (ResNet34) = 93.2 %, Accuracy (SVM)= 86.8 %, Accuracy (RF)= 77.7 %	[98]
MPs from six industrial wastewater treatment plants and the Main and Nidda rivers in Germany	Analysis of MPs in 47 environmental or wastewater samples	Raman	Classification	CNN	Accuracy > 90 %	[124]
MPs from sewage seal and foil, car tire, glove	Effectiveness in predicting emissions of MPs additives	LDIR	Regression	NNR	$R^2 = 98.5 \%$	[125]
MPs from wetland and reservoir in Guangzhou	Segmentation of MPs shapes	Biomicroscope	Image segmentation	UNet, UNet2plus	Accuracy (UNet)= 91.45 $\pm$ 5.93 %, Accuracy (UNet2plus) = 91.08 $\pm$ 6.18 %	[126]
MPs from daily supplies	Shape segmentation and classification of MPs for daily supplies	Scanning electron microscope	Image segmentation, Image classification	VGG16, UNet, MultiResUNet	Classification accuracy (VGG16) = 98.33 %, Segmentation IoU (UNet)= 78.37 %, Segmentation IoU (MultiResUNet)= 73.20 %	[127]
MPs in seawater	MPs detection and counting	Camera, microscope	Object detection	Faster-RCNN	Accuracy= 79.2–98.3 %	[123]
MPs from plastic bottles, garbage bags, crisper box, plastic wrap	Identify and analyze each component in the SERS spectra of six common MPs mixture	SERS	Classification	CNN, RF, SVM, KNN	Accuracy (CNN)= 99.5 %, Accuracy (RF)= 89.7 %, Accuracy (SVM)= 91.9 %, Accuracy (KNN)= 88.4 %	[128]
MPs in wastewater	Classification of municipal wastewater treatment plants in four forms: fibers, films, debris and particles	Samsung Galaxy S22 Android smartphone, stereomicroscope, FTIR	Image classification	Efficientnet_b7, Inception_v3, Resnet_v2_50, Resnet_v2_101, Mobilenet_v3	Accuracy= 98 %, 98 %, 97 %, 94 %, 98 %	[129]
MPs from oil in the ocean	Classification of MPs in ocean oil	SEM	Image classification	ResNet50	Accuracy= 99.86 %	[130]
MPs from dust-proof nets in Beijing	Estimation of the number of MPs based on the area covered by dust-proof nets and the abundance of MPs	Stereomicroscope, FTIR	Image segmentation	UNet	No model-specific performance metric values were provided	[131]
MPs from the beach of Playa del Poris in Tenerife Island, Spain	Shape segmentation of MPs particles and then MPs classification based on shape	Digital camera, Mobile phone	Image segmentation, Image classification	UNet, VGG16	Segmentation accuracy (UNet)= 98.82 %, Classification accuracy (VGG16)= 98.1 %	[121]
MPs in marine environment	Detection of 6 MPs particles in marine environment	Fluorescence microscope	Object detection	Faster-RCNN	Accuracy= 99 %	[132]

learn and extract relevant feature information directly from the input data. Avoiding the tedious step of manually processing data saves time and cost [98]. When processing MP data, the special model structure of CNN enables it to learn and extract and fuse complex feature information in MPs data. In addition, CNN are also highly adaptive in dealing with scenarios such as the shape, size, and orientation of MPs. Meanwhile, the pooling layer performs pooling operations on the feature maps generated by the convolutional layer to reduce the redundant output of the model and retain the most useful feature information. The pooling layer usually performs pooling operations on small local regions of the feature map and aggregates or selects the maximum or average values in these regions. This pooling operation effectively reduces the model complexity, improves the inference speed, and also preserves the global and local feature information to some extent. Fig. 6(A) illustrates the model structure flowchart of a CNN processing classification data of MPs. In the final layer of the model structure, the classification network employs the Softmax function as the loss function, which is commonly used in classification tasks. In fact, it is a supervised algorithm generated by logistic regression in a neural network that shows excellent performance in multi-class classification problems [100], and liquid-solid frictional nanogenerator for liquid MPs detection [128]. It is usually placed in the last layer of the neural network, and the final output results in probabilistic information for multiple classes. The regression prediction model structure of CNN is highly similar to the classification prediction model structure, with the difference lying in the last layer of the regression prediction task, which employs the Linear function. The Linear function fits multiple input features to obtain specific regression output values.

CNN are applied to numerical data classification or regression and it is able to capture complex nonlinear relationships and perform well on large-scale datasets. However, the training time is long and prone to overfitting. Therefore, CNN are used for numerical data classification or regression requires large-scale data and adequate computational resources.

4.2. Image classification

Previous researchers often used HSI technology to observe, but it lacks clear MPs shape classifications, such as fibrous, rod-shaped, elliptical, spherical, quadrilateral, triangular, free-form, and unidentifiable shapes [134]. However, due to the countless forms of MPs, dynamic environmental forces, and methods for characterizing MPs being both time-consuming and expensive, understanding the transportation of MPs in the environment is challenging [135]. The use of DL technology for image classification of MPs is of great significance. The shape, color, and edge smoothness of MPs are the key factors in understanding the transportation of MPs in the environment. These appearance factors of MPs are mainly in the form of images. The CNN method processes the RGB image information of MPs, different shapes represent different pixel points and different colors represent different number of channels (the images are all with R (red), G (green), B (blue)), CNN performs feature extraction with different pixel values and number of channels and finally matches these features with human given category labels to achieve high prediction accuracy. Fig. 6(B) shows a schematic diagram of CNN image classification, which uses a series of convolution operations to extract multi-level features of input images, and finally classifies the proposed features through fully connected operations to obtain the final category information. Shi et al. (2022) [127] imaged MPs of various chemical compositions and shapes from daily necessities using scanning electron microscopy, manually annotated 237 MPs particles (fragments or beads) within the range of 50 μm - 1 mm and fibers with a diameter of approximately 10 μm , and trained DL for MPs classification using VGG16 neural network based on their shapes, achieving an accuracy rate of 98.33 %. Akkajit et al. (2023) [129] constructed five CNN network models to classify four types of MPs from municipal wastewater treatment plants: fibers, films, debris, and particles, all of which achieved higher than 90 % accuracy in image classification.

CNN image classification methods show excellent performance in determining the type and shape of individual MPs. However, it is limited when dealing with images of mixtures.

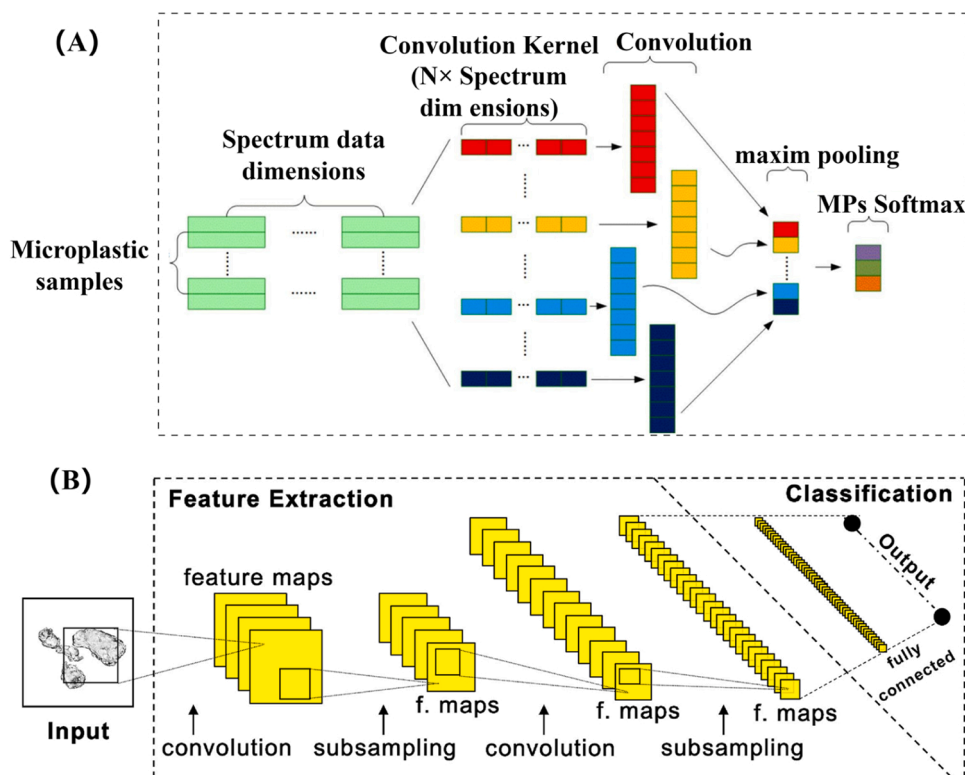


Fig. 6. Schematic diagram of CNN structure. (A) CNN for MPs classification and regression [133]; (B) CNN for MPs image classification [134].

4.3. Image detection

In response to the shortcomings of CNN in handling the classification of MPs images, object detection methods can handle images containing various types of MPs. Object detection can identify and annotate multiple categories on a single image, with the object detection model exhibiting fast inference speed and accurate detection accuracy, enabling simultaneous selection of multiple targets on an image in a short time. Models from the YOLO [136] series are widely used in the field of object detection due to their clear model structure, high stability, and strong detection capabilities. As shown in Fig. 7(A), YOLO consists of three parts: Backbone mainly extracts features from input images, Neck is responsible for multi-scale feature fusion, and Head makes final regression predictions to output prediction results. Grekov et al. (2022) [15] using the YOLOV5 detection model to detection and recognition of small objects in the marine environment. Microplankton and MPs samples were prepared, and a classified image database was collected based on these samples for training and testing the object detection mode. Fig. 7(A) shows the detection results of YOLOV5 for small objects, with the numbers representing the similarity to the true class (0 for completely dissimilar, 1 for completely similar).

Object detection is a mature technique in DL field, which uses DL structure to detect small MPs quickly and efficiently. Some typical target detection algorithms such as YOLO, Faster-RCNN [132] etc. can detect small target MPs and achieve high accuracy rate. However, these algorithms require a large amount of training data, and these algorithms require high computational resources on hardware. Therefore, the

adoption of these algorithms requires high-quality datasets and computationally powerful devices.

4.4. Image segmentation

Object detection only allows for bounding box annotations of target categories in images. Image segmentation techniques in DL, building upon object detection, perform more complex tasks. Image segmentation can segment target categories based on edge shapes and annotate category information [124]. Fig. 7(B) shows the workflow of the UNet [137] model segmenting MPs data in water bodies. Firstly, water samples are collected and preprocessed, then experimental samples are observed under a microscope to obtain MPs image datasets. Image datasets are annotated using labeling tools and split into training and validation sets at a certain ratio. The training and validation sets are fed into UNet and improved models for model training and validation. After saving the trained model weights, a small amount of the test set is used to test the MPs segmentation model, obtaining the final segmentation mask image. Image segmentation techniques perform pixel-level segmentation of MPs edge shapes, and the resulting mask images better suit downstream tasks, such as automatic counting of MPs particles on beaches [122], and instance segmentation of dense, tangled fibers for further quantification [127]. The combination of image segmentation models with object-aware pixel embeddings, using trained models, enables high-precision segmentation and classification of MPs microscope images in a short time.

Image segmentation techniques can not only distinguish and localize

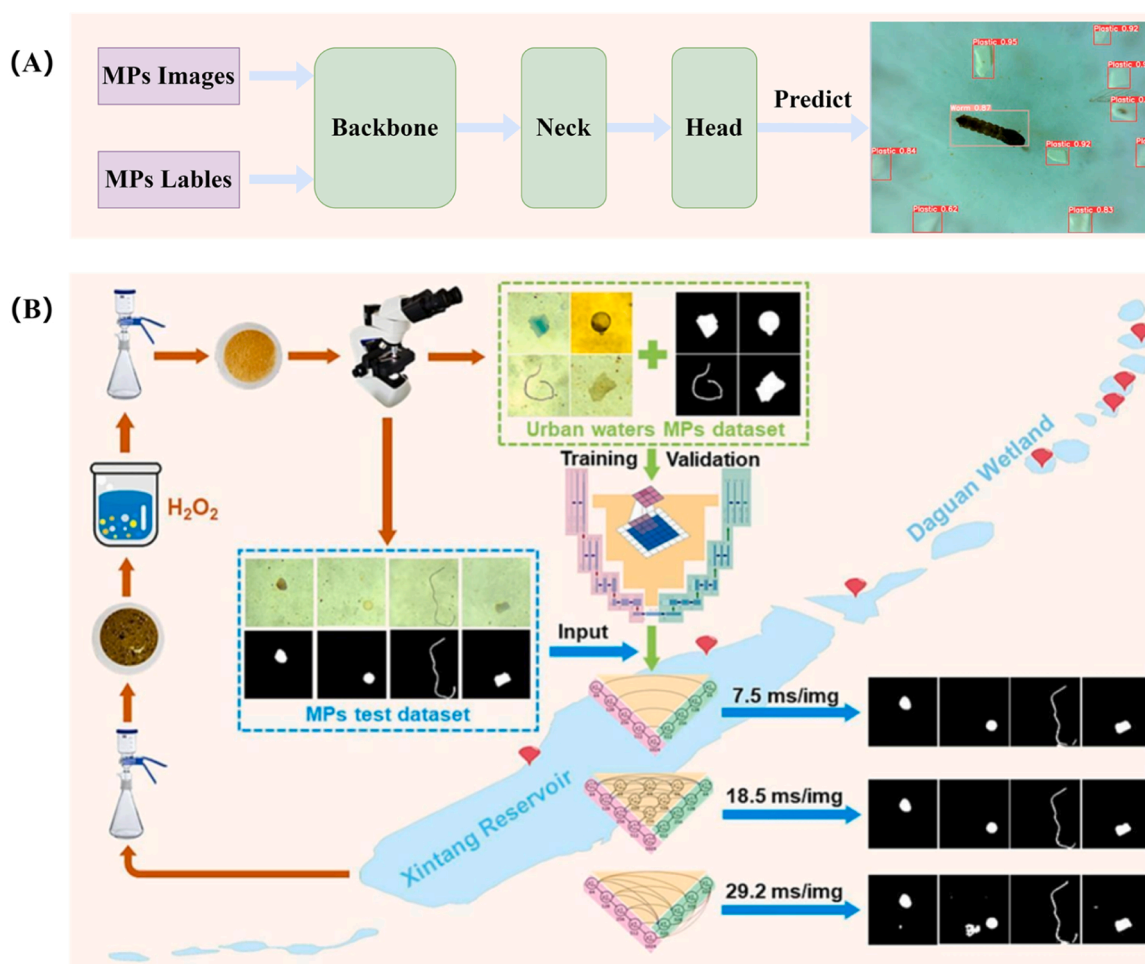


Fig. 7. Schematic of object detection and image segmentation. (A) Schematic diagram of MPs object detection [15]; (B) Schematic diagram of MPs image segmentation [126].

targets by category, but also segment the edge shape of the target at the pixel level, integrating the advantages of image classification and target detection. However, the technique also faces strong computational resource constraints and also requires a large amount of data for training.

5. Challenge and future prospect

AI-driven acceleration of MPs to recognize and predict environmental behaviors has been shown to have application in existing work. However, ML has numerous challenges in the path of prediction of MPs.

Data. Currently, there are few publicly available MPs datasets, and usually only a small sample of datasets is available experimentally. A good predictive model requires a large amount of data for training. In vision tasks, especially object detection and image segmentation, each image needs to be cleaned to eliminate blurred, mutilated, unbalanced and insufficient number of categories, etc., and labeling work needs to be performed on each image. The small amount of data and poor data quality lead to insufficient model robustness, which hinders the development of ML for MPs applications.

Algorithms. Different algorithms should be chosen for different MPs data types and different tasks, a good choice of algorithm can make the prediction more accurate. However, how to choose a good algorithm becomes a tricky problem. It is crucial to choose a good algorithm to be used as a baseline, to modify the model structure in the baseline, and to improve the prediction performance of the model to improve the generalization ability. Meanwhile, the evaluation indexes for model training are not rich enough to assess the generalization ability of the model in all aspects. Although ML algorithms have strong prediction ability in many fields, the models usually lack interpretability, which is crucial for researchers to understand and trust the models.

Tasks. In most of the current MPs prediction work, the main tasks are classification and regression of numerical data, classification, detection, and segmentation of picture data, etc. The number of tasks is small and cannot support all the MPs prediction. MPs is commonly analyzed by methods such as FTIR and Raman spectroscopy, which extract data as sequences, yet no work has been found to predict MPs sequences. Meanwhile, 1D data (numerical data) and 2D data (image data) do not fully characterize MPs, and more 3D data should be investigated.

Tools. ML algorithms provide a reliable solution for MPs prediction. However, modeling usually requires a rich theoretical foundation and programming skills, and modeling by non-experts is inefficient and difficult. There is an urgent need for a reliable platform or tool that integrates almost most of the classical algorithms. Without writing code, model evaluation metrics and prediction results can be obtained directly, which greatly reduces the time cost of researchers.

In summary, AI-driven prediction of MPs faces multiple challenges from data, algorithms, tasks, and tools, and requires reliable solutions. In the future, about ML-driven methods for analyzing MPs remain continuously popular. The following are some potential research directions:

- (1) Building large datasets, including multiple types (such as numerical data, images, sequences, videos, etc.) from multiple sources (soil, water, etc.) for research;
- (2) Expanding more MPs datasets and using DL-based GAN [138] models for data cloning avoided overfitting for model training;
- (3) Development of additional reliable evaluation indicators to provide a reliable interpretation of the model;
- (4) Design more high-performance and low-computational-complexity algorithms (such as LeaNet [139], MALUNet [140]) to improve task prediction capabilities for better deployment on mobile devices with limited computational resources;
- (5) Conduct more task studies, sequences (such as Transformer [141], LSTM [142], GRU [143]), text (such as ChatGPT4), 3D image (such as VMNet [144]);

- (6) Build a 0-code ML platform which provides faster solutions for non-specialists.

6. Conclusions

AI-driven prediction of MPs and behavioral analysis of MPs' environments have been proven to be reliable methods. In particular, ML, which can process the diversity of MPs' feature information, has realized the efficient prediction of MPs for various environments. ML-based DL methods stand out in the analysis of numerical data and picture data. In this paper, we meticulously analyze various data sources of MPs and introduce different data preprocessing methods for different data types. Numerous algorithmic features of traditional ML such as DT, RF, SVM, KNN, linear regression, gradient boosting regression are presented respectively, including their advantages and disadvantages. The application of DL on various tasks of MPs such as numerical data classification and regression, image classification, target detection, image segmentation is also presented separately. In addition, this review develops an analysis of the challenges of applying ML algorithms on MPs, while suggesting some potential future directions. We hope that this review will provide some insights for future research on MPs.

Environmental implication

Numerous studies have demonstrated the detrimental effects of microplastics as environmental pollutants on aquatic ecosystems and soil biota. Leveraging machine learning and deep learning for microplastics investigation facilitates long-term and large-scale monitoring of microplastics concentrations, types, and distribution. This study comprehensively reviews the applications of machine learning and deep learning in various domains related to microplastics, including classification, regression prediction, objective detection, and image segmentation. Additionally, the proposed future directions offer insights for the development of more intelligent approaches to investigate the environmental behavior of microplastics.

CRediT authorship contribution statement

Ying Sun: Investigation, Data curation. **Yueyue Dai:** Software. **Hongfang Yu:** Software, Data curation. **Daji Ergu:** Data curation. **Ming Wang:** Visualization, Formal analysis. **Pan Zhou:** Supervision, Resources, Project administration, Funding acquisition. **Binbin Hu:** Writing – original draft, Software, Methodology, Conceptualization. **Hai Zhou:** Validation. **Yaodan Dai:** Writing – original draft, Visualization, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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